Square sections, alternating assemblies and free extensions

Human Compression Project

Abstract

Compatible products give regular polytopes with any prescribed sequence of two-power square sections. Coordinate kernels determine their exact orders, flatness and deletion properties. A normal-kernel recognition theorem then identifies their complete presentations, while jointly faithful deletions identify the alternating two-branch construction. The same application also recovers the three free-extension results through the torsion theorem for amalgams. These are the six principal construction and extension theorems of Cunningham, Feng, Hou and Schulte. The geometric assembly and free-extension identifications are retained in explicit forms; their algebraic consequences are proved here, including infinite inputs.

1 Marked groups and geometric inputs

A string C-group is generated by nonidentity involutions $r_0, \dots, r_{d-1}$, satisfies $[r_i, r_j] = 1$ for $|i-j| > 1$, and has the intersection property

$$G_J \cap G_K = G_{J\cap K}, \qquad G_J = \langle r_j : j \in J\rangle.$$

We retain the regular-polytope correspondence stated precisely in [UA]: such a marking gives a regular abstract polytope of rank d, with the group acting regularly on flags. Its consecutive rank-three section on labels $i, i+1, i+2$ has that marked subgroup as its group. A face of rank j has stabilizer $G_{I\setminus\{j\}}$. Faces are incident when their stabilizer cosets meet. These standard correspondences carry the original attribution [MS02].

A deletion keeps a specified subset of the markers and sends all other markers to the identity. UA.D proves that a family of all such deletions gives the intersection property and

$$G = \langle\!\langle r_i : i \notin J\rangle\!\rangle \rtimes G_J.$$

At an initial or final cut, this split quotient is the flat amalgamation property (FAP) used in [CFHS25]. We use this algebraic characterization. Flatness is a different assertion: a polytope is (j,k)-flat if every j-face is incident with every k-face. It will follow from an actual factorization by face stabilizers, not merely from deletion maps.

Write H_n, $n \geq 5$, for the affine square group in UA.Square. Its order is 2^n, its adjacent product orders are $4, 4$, and its endpoint product order is 2. It has every Boolean deletion. Its complete affine model is

$$H_n = (\mathbb{Z}^2 \rtimes D_8)/\Lambda_n, \qquad \Lambda_n = \begin{cases} 2^e\mathbb{Z}^2, & n = 2e+3, \\ 2^e\langle(1,1),(1,-1)\rangle, & n = 2e+4, \end{cases} \quad e \geq 1.$$

The full signed-permutation coordinate D_8 is retained even in a small quotient on which its action is not faithful. The natural map $H_n \to H_5$ has kernel

$$N_n = \gamma_3(H_n) = 2\mathbb{Z}^2/\Lambda_n, \qquad |N_n| = 2^{n-5}. \tag{1}$$

It is injective on every proper marked subgroup: the three pairs retain their exact orders and all singleton subgroups retain order two.

For completeness, UA.SquareRecognition proves that every finite two-group string C-group of type $\{4,4\}$ is exactly one of these marked groups. Its full proof uses orientation parity, the torsion-free translation kernel of the square reflection presentation, and the Euclidean algorithm in $\mathbb{Z}[i]$. Under the regular-polytope correspondence this gives the two square-map families in the source's Lemma 4.1, one for each $n \geq 5$. Their group orders are $8s^2$ and $16s^2$, respectively, with $s = 2^e$. Thus the classification here follows from a complete shared proof, rather than an unproved exhaustion of the affine examples. The finite construction below itself needs only the explicit groups H_n.

2 One construction for every sequence of square sections

Let T be a finite forest with vertex set I. The coordinate group B_T of UA.Forest has binary coordinates on the vertices and edges. After choosing an orientation for each edge, multiplication is

$$(v,z)(w,t) = (v+w, z+t+\beta(v,w)), \qquad \beta_{i\to j}(v,w) = v_j w_i.$$

Its order is $2^{|I|+|E(T)|}$; deleting coordinates gives every Boolean deletion. Each two-edge path has marked group H_5. Choose distinct two-edge paths S and parameters $n_S \geq 5$. Attach $H_{n_S} \to (B_T)_S = H_5$ on each support. These maps are faithful on proper marked subgroups by (1), and the supports form an antichain. UA.SK and its local, deletion and lower-central clauses give the complete compatible product G, with its actual marker tuples, and

$$|G_J| = 2^{|J|+|E(T[J])|+\sum_{S\subseteq J}(n_S-5)}, \tag{2}$$

$$\gamma_3(G) = \prod_S N_{n_S}, \qquad G/\gamma_3(G) = B_T. \tag{3}$$

It has all Boolean deletions and exact local groups $G_S = H_{n_S}$. The order in (2) is the order of the group generated by the marker tuples: generation of the whole compatible product is part of UA.SK.

Theorem 1 (Cunningham–Feng–Hou–Schulte, Theorem 5.1)**.** *Let $d \geq 3$ and $n_3, \dots, n_d \geq 5$. There is a regular rank-d polytope $\mathcal{P}$ whose consecutive square sections have groups $H_{n_3}, \dots, H_{n_d}$, with*

$$|\Gamma(\mathcal{P})| = 2^{\sum_{j=3}^{d} n_j - 3(d-3)}. \tag{4}$$

For $d \geq 4$ it is $(d-4, d-1)$-flat, hence flat. It has the FAP with respect to its co-$(d-4)$-faces, co-$(d-3)$-faces and facets. At $d = 3$ the co-(-1)-face cut means the whole polytope.

Proof. Take the path $0, 1, \dots, d-1$ and attach $H_{n_{i+3}}$ on every $S_i = \{i, i+1, i+2\}$, $0 \leq i \leq d-3$. Distant markers commute in the base and in every local factor, so the resulting IP marking is string. Its generators have exact order two by their base images. Local recognition gives the prescribed marked square sections. Equation (2), with $J = I$, gives

$$|G| = 2^{2d-1+\sum_{j=3}^{d}(n_j-5)} = 2^{\sum_{j=3}^{d} n_j - 3(d-3)}.$$

The regular-polytope correspondence now gives the stated object and order.

All deletions exist. In particular delete onto $\{d-3, d-2, d-1\}$, onto $\{d-2, d-1\}$, and onto $\{0, \dots, d-2\}$. UA.D supplies the three split normal-closure decompositions, which are exactly the stated FAP assertions. For $d = 3$ the first deletion is the identity, so its interpretation needs no exception to the algebraic statement.

For flatness assume $d \geq 4$, and put $L = \{0, \dots, d-2\}$ and $M = \{d-3, d-2, d-1\}$. Every support is contained in L or M. The vertex and edge coordinates give $B_T = (B_T)_L (B_T)_M$: one can first choose the two factors' vertex coordinates, then assign each central edge coordinate to a side containing that edge, correcting the product's bilinear contribution there. UA.SK.Separator gives $G = G_L G_M$. The

first factor is a facet stabilizer; the second lies in the stabilizer of the base $(d-4)$-face. Thus their full stabilizers multiply to G. Any right coset of one meets any right coset of the other, so every such face is incident with every facet. Every vertex lies in a $(d-4)$-face; it is therefore incident with every facet as well. This proves the two flatness conclusions. □

The group is also precisely the particular iterated mix used in the source, not merely another group with its order and sections. Projection into $\prod_i H_{n_{i+3}}$ is injective: an element with all local coordinates trivial has every triple deletion of its base coordinate trivial, and the triples cover every vertex and edge coordinate of B_T. Its image is generated by the marked tuples in those factors. Equation (8) of the source and its induction in Theorem 5.1 produce exactly these tuples, with only reassociation of the direct product. The marked groups are consequently isomorphic.

3 Recognizing the complete universal presentation

Let P_0 have the string Coxeter relations of type $\{4, \dots, 4\}$ and the complete lattice relations of $H_{n_{i+3}}$ on every consecutive triple. Let P be its quotient by

$$[(r_i r_{i+1})^2, (r_{i+2} r_{i+3})^2] = 1 \quad (0 \le i \le d-4). \tag{5}$$

The following local identity supplies the required normalization; its proof is common to every rank and parameter.

Lemma 2. *If involutions a, b, c, d satisfy $[a,c] = [a,d] = [b,d] = 1$ and $[(ab)^2, (cd)^2] = 1$, then d commutes with both $(cbab)^2$ and $(abcb)^2$.*

Proof. Put $x = bab$, $A = (ab)^2 = ax$, and $F = (cd)^2$. Since a commutes with F, the relation $[A, F] = 1$ gives $[x, F] = 1$. Also $[x, d] = 1$, hence $[x, Fd] = [x, cdc] = 1$. Conjugating by c gives $[cxc, d] = 1$. Both factors of $(cbab)^2 = (cxc)x$ commute with d. Finally $abcb = b(cbab)^{-1}b$, and b commutes with d. □

Theorem 3 (Cunningham–Feng–Hou–Schulte, Theorem 5.3)**.** *The group P_0 is the universal regular-polytope group with the indicated consecutive square sections. The group constructed in Theorem 1 is precisely its marked quotient P by (5).*

Proof. Let H denote the group of Theorem 1. Its local sections have the exact complete presentations used in P_0. Relations (5) hold in its class-two base. In each local triple coordinate at least one of the two disjoint edges loses an endpoint, so that pair product has square one. Thus every relator of P holds in H, giving a marked surjection $P \to H$. Each local square group maps to P by the defining relations; projection onward to its own coordinate of H is the identity, so this local map is faithful.

Let K_i be the image of $\gamma_3(H_{n_{i+3}})$ in P. It is generated by the two doubled translations appearing in the lemma. The three local markers normalize it. The lemma and its reversed version show that the nearest outside markers centralize its generators; every farther marker commutes with all local markers by the string relations. Hence each K_i is normal in P.

Kill all the K_i. Every consecutive triple becomes class two. Every adjacent square therefore commutes with its endpoints and its nearest outside markers; the string relations give commutation with all remaining markers. The quotient is class two and satisfies the complete path-base presentation of UA.Forest. Conversely its map to H followed by the base map gives the marked surjection onto that same path base. The two maps are inverse on generators, so this is the exact residual quotient. All four hypotheses of UA.Recognition now hold. It identifies P with H without assuming P finite or estimating its order.

Finally every defining relator of P_0 survives all deletion assignments. Even Coxeter relators do so directly; UA.Square proves the assertion for each complete local lattice relation. Thus P_0 has IP by UA.D. Its generators are nonidentity involutions, and each local group embeds because its map onward

to H is faithful. Hence P_0 is a string C-group with the required exact sections. Every other such marked group satisfies all of its relators and is its marked quotient. The standard covering correspondence gives the universal regular polytope. No finiteness of P_0 is used or asserted. □

In the smallest diagonal square group H_6, the two doubled translations coincide. Formula (1) uses their full lattice subgroup and does not assume that their two cyclic subgroups are independent. This preserves the previously recorded qualification to the source's presentation proof.

4 Two branches and the full alternating automorphism group

We retain the following precise FAP case of the alternating-assembly theorem of Monson and Schulte [MSA, Theorem 5.3]. If regular $(d-1)$-polytopes $\mathcal{P}, \mathcal{Q}$ have an identified common facet $\mathcal{K}$ and FAP decompositions

$$\Gamma(\mathcal{P}) = N_P \rtimes \Gamma(\mathcal{K}), \qquad \Gamma(\mathcal{Q}) = N_Q \rtimes \Gamma(\mathcal{K}),$$

then the tail-triangle group $(N_P \times N_Q) \rtimes \Gamma(\mathcal{K})$, with the two distinct final markers, gives an alternating semiregular d-polytope with those facets. Its full automorphism group is that group when the two facets are not isomorphic, and its extension by the branch-swapping C_2 when they are the same marked polytope. In the latter case the polytope is regular. This geometric and full-automorphism identification is retained; group IP alone would not establish it.

Theorem 4 (Cunningham–Feng–Hou–Schulte, Theorem 6.1)**.** *For $d \geq 4$, $n_3, \dots, n_{d-1} \geq 5$ and $m_{d-1} \geq 5$, there is an alternating semiregular rank-d polytope with regular facet lists*

$$(n_3, \dots, n_{d-2}, n_{d-1}), \qquad (n_3, \dots, n_{d-2}, m_{d-1}).$$

Its full automorphism group has order

$$2^{\sum_{j=3}^{d-2} n_j + n_{d-1} + m_{d-1} - 3(d-3)} \tag{6}$$

when the last parameters differ, and twice this order when they agree. For $d = 4$ the shared list is empty and the common facet is a square.

Proof. Take the tree with common prefix $0, \dots, d-3$ and two final leaves u, v, both adjacent to $d-3$. Attach the two prescribed lists of square groups along the two paths, with the shared prefix supports counted once. The construction leading to (2) applies. Let L and M be the two full path label sets, and put $C = L \cap M$. Local recognition and the coordinate construction identify G_L, G_M with the two groups of Theorem 1. Their common subgroup G_C is its shorter path group, or the square dihedral group of order eight when $d = 4$.

The deletions onto L and M are jointly faithful. Every vertex and edge coordinate of the base is retained on at least one side, and every local kernel support lies entirely on a side, where its coordinate is retained. An element killed by both deletions therefore has every coordinate trivial. UA.Mix identifies the group with the full compatible product $G_L \times_{G_C} G_M$. This step uses joint coordinate detection and actual generation of the compatible product, not an order comparison. By UA.Fiber it is exactly $(\ker(G_L \to G_C) \times \ker(G_M \to G_C)) \rtimes G_C$, with the original tail-triangle marking. The retained assembly theorem applies.

The common group has order $2^{\sum_{j=3}^{d-2} n_j - 3(d-5)}$, including 8 for the empty list at $d = 4$. The two kernels have orders $2^{n_{d-1}-3}$ and $2^{m_{d-1}-3}$ by (2). Multiplication gives (6). Unequal last parameters give unequal facet-group orders, so their polytopes cannot be isomorphic. Equal parameters give the identical marked construction and its branch interchange. The full-group alternative in the retained theorem supplies exactly the factor of two asserted. □

The co-$(d-4)$-section not lying inside one of the two facet lists is not separately prescribed by this theorem. No extra statement about that section is used to prove (6).

5 Free extensions and two-primary torsion

Following the source, an infinite two-group means a finitely generated infinite group every finite-order element of which has two-power order. It need not be a torsion group. UA.Amalgam proves that this torsion property passes to subgroups, to extensions with finite two-group quotient, and to amalgamated free products. Its proof retains the precise amalgam normal-form theorem and proves infinitude when the common subgroup is proper in both factors.

We retain three geometric identifications, including existence, universality, rank and symmetry type:

1. A regular rank-d polytope $\mathcal{K}$ with facet $\mathcal{F}$ has a free regular extension of rank $d+1$, with group

$$\Gamma(\mathcal{K}) *_{\Gamma(\mathcal{F})} (\Gamma(\mathcal{F}) \times C_2).$$

   This is the standard regular free-extension construction [MS02,4D4].

2. An oriented chiral rank-d polytope $\mathcal{K}$ with directly regular facets $\mathcal{F}$ has a free oriented chiral extension of rank $d+1$, with group

$$\Gamma(\mathcal{K}) *_{\Gamma^+(\mathcal{F})} \Gamma(\mathcal{F}), \qquad \Gamma(\mathcal{F}) = \Gamma^+(\mathcal{F}) \rtimes C_2.$$

   Here the facet rotation subgroup embeds in $\Gamma(\mathcal{K})$. This is the free chiral-extension theorem of Schulte and Weiss [SW95].

3. Regular rank-d polytopes $\mathcal{P}, \mathcal{Q}$ with isomorphic facets $\mathcal{K}$ have a universal alternating semiregular extension of rank $d+1$. Its group is

$$\Gamma(\mathcal{P}) *_{\Gamma(\mathcal{K})} \Gamma(\mathcal{Q})$$

   when $\mathcal{P} \ncong \mathcal{Q}$, and the extension of this group by the branch-swapping C_2 when $\mathcal{P} \cong \mathcal{Q}$. This is the universal alternating construction of Monson and Schulte [MST12,MST21].

All input ranks in these statements satisfy the source's convention $d \geq 3$. The amalgamating subgroup is proper in each factor: in the polytope factor it is a proper face or facet-rotation stabilizer, while in the added factor of the first two constructions it has index two. These are the actual geometric subgroup embeddings, not abstract groups of the same order.

Theorem 5 (Cunningham–Feng–Hou–Schulte, Theorems 7.1–7.3)**.** *In each of the three identifications above, if the input automorphism groups are finite or infinite two-groups in the stated convention, the full automorphism group of the resulting extension is a finitely generated infinite group with only two-power finite torsion.*

Proof. For the regular extension, $\Gamma(\mathcal{F})$ is a subgroup of the input group and has the torsion property. Its product with C_2 has the property by the finite-extension clause of UA.Amalgam. For the chiral extension it is the rotation subgroup $\Gamma^+(\mathcal{F})$ that embeds in the input group. Its index-two extension $\Gamma(\mathcal{F})$ again has the property. This does not assume that the full facet group embeds in the chiral automorphism group. In the alternating case both factors are input groups and already have the required property.

In each case every finite-order element of the amalgam is conjugate into one of its factors, so it has two-power order. Properness on both sides provides reduced alternating words of arbitrarily large length and proves that the amalgam is infinite. All factors are finitely generated by their finite marked lists, so their amalgam is finitely generated. In the equal alternating-facet case, its additional extension by C_2 still has the torsion property and remains infinite and finitely generated. The retained geometric identifications supply the rank, symmetry and universal-extension conclusions in the three source theorems. □

6 Coverage and attribution

The source edition is arXiv:2512.15511v1, dated 18 December 2025. We have proved its Theorem 5.1(a–d), complete Theorem 5.3, Theorem 6.1(a–b), and Theorems 7.1–7.3, relative to the explicitly retained geometric statements. The full square-recognition theorem in the umbrella also recovers Lemma 4.1. The source's unnumbered remarks and power-polytope discussion are not included in this principal contract.

The finite constructions use the same supported-kernel theorem. Its normal-kernel recognition clause proves the complete presentation, and joint coordinate detection proves the full two-branch product. Local normalization and the base quotient remain explicit work. The infinite results use a different common mechanism, the amalgam normal form; they are not inferred from the finite construction. The original results and all earlier geometric inputs retain their stated attribution.

References

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[3] [MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.

[4] [MSA] B. Monson and E. Schulte, *The assembly problem for alternating semiregular polytopes*, *Discrete & Computational Geometry* 64 (2020), 453–482, Theorem 5.3, in the precise FAP form retained above.

[5] [SW95] E. Schulte and A. I. Weiss, *Free extensions of chiral polytopes*, *Canadian Journal of Mathematics* 47(3) (1995), 641–654.

[6] [MST12] B. Monson and E. Schulte, *Semiregular polytopes and amalgamated C-groups*, *Advances in Mathematics* 229 (2012), 2767–2791.

[7] [MST21] B. Monson and E. Schulte, *Universal alternating semiregular polytopes*, *Canadian Journal of Mathematics* 73 (2021), 572–595. The exact group and full-automorphism interface is stated above.